

Obs sumacion cesaro convolucion nucleo fejer

Observación 1. Como por la obs-sumas-parciales-serie-fourier-convolucion-nucleo-dirichlet/ob

$S_k f = f * D_k$, tenemos que $\forall N \in \mathbb{N}$:

$$\begin{aligned} \forall x \in \mathbb{R} : \sigma_N f(x) &= \frac{1}{N} \sum_{k=0}^{N-1} f * D_k(x) = \frac{1}{N} \sum_{k=0}^{N-1} \int_0^1 f(x-t) D_k(t) dt \\ &= \int_0^1 f(x-t) \left(\frac{1}{N} \sum_{k=0}^{N-1} D_k(t) \right) dt = f * \left(\frac{1}{N} \sum_{k=0}^{N-1} D_k(\cdot) \right) (x) = f * F_N(x). \end{aligned}$$